

ZERO-TO-JOB-READY DATA ANALYST 2026

M02 - EXCEL FOR ANALYSTS

Beginner-first analyst Excel | 11 lessons + defended project

LEARN -> WATCH -> FOLLOW -> DO ALONE -> VALIDATE -> EXPLAIN -> MASTER

Technical difficulty rises; the teaching voice stays beginner-first.

This book is your teacher

- You already know basic Excel from M01, but every new analyst concept is still introduced from first principles. A new formula name is never treated as self-explanatory.
- Videos are used as visual demonstrations, not as hidden prerequisites. Watch only the mapped segment, return immediately to Excel, reproduce the action, then do a changed version.
- Protected reviews remain closed until a genuine attempt is saved. After review/repair, close the answer material and complete a fresh changed retry.
- Your goal is not a pretty workbook. Your goal is a workbook whose grain, formulas, transformations, Pivots, lookups, report and validations you can defend.
- There is no forced 25-minute timer. Course Mentor study sessions will use manual START STUDY / END STUDY real elapsed time.

Normal lesson loop

1. Understand why + mental model
2. Learn new words
3. Watch exact visual if assigned
4. Follow teacher demo once
5. Do guided workbook task
6. Close support and do changed task
7. Validate/reconcile
8. Explain back + save evidence

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Module prerequisite and tool map

0. Prerequisite

M01 Excel Fundamentals: safe rectangular data, formula basics, five basic functions, relative/absolute references, real Excel Tables, safe sort/filter, basic charts, validation and explain-back. If these are not independent, repair M01 before treating M02 as normal progression.

Tools / files

- C1_M02_EXCEL_FOR_ANALYSTS_CANONICAL_PRACTICE_WORKBOOK_RC1.xlsx - guided labs, source tables, project outputs, Evidence_Log and Mastery_Tracker.
- Excel desktop. Power Query and Data Model/Power Pivot features vary by Excel edition; the course teaches the conceptual competency even when a particular UI is unavailable.
- C1_M02_EXCEL_FOR_ANALYSTS_PROTECTED_PRACTICE_REVIEW_RC1.pdf - closed until a saved attempt.
- Gate A/B/C workbooks and separate protected reviews - B/C only after the prior Gate fails and repair is completed.

Important boundary

- M02 teaches analyst-strength Excel, not every obscure Excel feature.
- Power Query is introduced before you are asked to use it in the project or Gate.
- Relationship/cardinality reasoning is taught before relationship use is assessed.
- Advanced BI/DAX belongs later; this module does not smuggle those skills into independent tasks.

Exact practice, review, project and Gate route

- Study EFA1. Use the assigned segment only if useful, then reproduce the teacher example in P-EFA1_IF.
- Complete the guided task. Close the lesson and do the changed fresh task. Validate and append evidence.
- Open the protected practice review only after the saved attempt, and only when you need to compare/repair.
- Repeat EFA1-EFA11. Use My_Pivots and My_Report for EFA8/EFA9; those output sheets now physically exist.
- Complete M-XLA: Prepared_Data -> Analysis_Pivots -> Management_Report -> Validation_Log. Defend the project before Gate A.
- Attempt Gate A independently. A clean competency pass ends the Gate.
- Gate B requires failed A + Review A + targeted repair + explicit assignment. Gate C requires failed B + Review B + targeted repair + explicit assignment.

Critical-failure rule

Wrong grain, hidden errors/not-found values, unsafe key handling, unreconciled results, broken refresh/repeatability or inability to explain the work can block a pass even if the workbook looks polished. No percentage-average shortcut.

EFA1 - IF, IFS and Logical Tests

Practice: P-EFA1_IF

1. Why this lesson matters

Analyst Excel work is judged by trustworthiness and repeatability, not by how many features are used. Every transformation, formula, Pivot/query and visual must preserve meaning and survive a changed input.

2. What you are learning

IF, IFS and Logical Tests at working independence: perform the analyst task, validate it, explain why the logic is safe, and recover from one realistic failure.

3. What you already need / prerequisite

M01 Excel Fundamentals only. IF/IFS/AND/OR are introduced here from zero.

4. Picture it in your head

Business rules must become TRUE/FALSE tests without changing their meaning. IF handles two branches; IFS handles ordered bands; AND/OR combine conditions. Boundary cases are part of the rule, not an afterthought.

5. New words before we use them

Logical test = expression returning TRUE/FALSE. Branch = output chosen by a rule. Boundary = exact cutoff such as 90 or due date. AND = all conditions true. OR = at least one condition true.

6. Watch / visual source

RECOMMENDED. 5 Advanced Tips on how to use IF formula in Excel that will SURPRISE you | 0:00-4:45. Why watch: See simple IF, nested/IFS, then the multi-condition section that supports AND/OR logic before reproducing the workbook task. Internal fallback: This lesson plus P-EFA1_IF fully teach the rule/boundary logic.

[Open mapped source](#)

EFA1 - Follow with me once

7. Follow with me once - fully worked teacher demonstration

Promised Date E2 and Actual Date F2: =IF(F2<=E2,"On Time","Late"). For score bands: =IFS(D2>=90,"A",D2>=75,"B",D2>=60,"C",TRUE,"D"). Test exactly 90, 75 and 60.

8. What you should see / be able to say

The formula matches the written business rule, boundary cases are tested manually, missing/cancelled cases are deliberately handled, and copied formulas remain row-correct.

9. Why it worked

The action matches the written business meaning and keeps a validation path back to source values. The learner is not asked to trust a formula, Pivot, query, relationship or visual merely because Excel produced an output.

10. Common beginner mistakes

Testing only easy cases; using > when the written rule says >=; deeply nesting IF before writing the business rule; letting blanks silently become a category.

11. If you get stuck - recovery ladder

If the formula becomes deeply nested, rewrite the rule in plain language and use helper columns or IFS. If exactly-at-cutoff cases fail, inspect >= versus > and rule order. Recovery ladder: 1) stop editing; 2) restate the business rule/grain; 3) locate the exact formula/menu/query/relationship; 4) inspect the earliest wrong step; 5) reveal only a structural hint; 6) use protected review only after a saved attempt.

EFA1 - Do alone, validate and master

12. Now you do a small version alone

On a fresh commission table, classify Commission Eligible only when Region = East AND Net Sales >= 50000. Test East at 49999, 50000 and non-East at 60000.

13. Validate your result

The formula matches the written business rule, boundary cases are tested manually, missing/cancelled cases are deliberately handled, and copied formulas remain row-correct.

14. Teach it back in your own words

Without the lesson open, explain if, ifs and logical tests in 60-90 seconds. State the business purpose, source/result grain, one validation, one common failure and why your handling preserves meaning.

15. Protected review + changed fresh retry

Save the genuine attempt in Evidence_Log first. If repair is needed, open the protected EFA1 review, compare only the weak competency, close the review, and complete a changed retry. Append the retry; never overwrite the failed/partial attempt.

16. Evidence / mastery + exact next route

Save workbook/query output, formula/range/query evidence, independent validation or reconciliation, one failure/recovery note, confidence 1-5, and a concise own-words explanation. PASS with evidence -> continue. PARTIAL/UNCERTAIN -> targeted repair + fresh changed retry. FAIL -> repair only the named weak competency before continuing.

EFA2 - SUMIFS, COUNTIFS and AVERAGEIFS

Practice: P-EFA2_ConditionalAgg

1. Why this lesson matters

Analyst Excel work is judged by trustworthiness and repeatability, not by how many features are used. Every transformation, formula, Pivot/query and visual must preserve meaning and survive a changed input.

2. What you are learning

SUMIFS, COUNTIFS and AVERAGEIFS at working independence: perform the analyst task, validate it, explain why the logic is safe, and recover from one realistic failure.

3. What you already need / prerequisite

M01 Excel Fundamentals plus the previous M02 lesson(s).

4. Picture it in your head

Conditional aggregation answers reusable business questions by applying criteria to aligned ranges. The result is only trustworthy when criteria types and row ranges match the source.

5. New words before we use them

Sum range = values to add. Criteria range = field being tested. Criteria = condition. Cell-driven criteria = report input cell instead of hard-coded text. Date window = $\geq$ start and $\leq$ end on the same date field.

6. Watch / visual source

RECOMMENDED. Best Excel Formulas in an hour - Beginner to PRO Masterclass | 1:51-8:28. Why watch: See COUNTIFS, SUMIFS and AVERAGEIFS with multiple criteria before building a cell-driven analyst report. Internal fallback: This lesson plus P-EFA2_ConditionalAgg fully teach and validate the formulas.

[Open mapped source](#)

EFA2 - Follow with me once

7. Follow with me once - fully worked teacher demonstration

Total selected Region + Category: =SUMIFS(Sales,Region,\$J\$7,Category,\$J\$8). Count selected Region above threshold: =COUNTIFS(Region,\$J\$7,Sales,">="&\$J\$11). Validate by filtering.

8. What you should see / be able to say

All criteria/sum/average ranges cover identical source rows; text/date/number criteria match stored types; one result reconciles to a filtered/manual check.

9. Why it worked

The action matches the written business meaning and keeps a validation path back to source values. The learner is not asked to trust a formula, Pivot, query, relationship or visual merely because Excel produced an output.

10. Common beginner mistakes

Criteria ranges do not cover the same rows; dates are text while criteria are dates; hard-coding criteria inside every formula; trusting zero without checking stored values.

11. If you get stuck - recovery ladder

If a result is zero unexpectedly, inspect spaces/case/data types and criteria operators. If ranges differ in size, correct the design rather than expanding blindly. Recovery ladder: 1) stop editing; 2) restate the business rule/grain; 3) locate the exact formula/menu/query/relationship; 4) inspect the earliest wrong step; 5) reveal only a structural hint; 6) use protected review only after a saved attempt.

EFA2 - Do alone, validate and master

12. Now you do a small version alone

Build a changed delivery report using two criteria and a date window. Change one report input and prove the answer updates without editing the formula.

13. Validate your result

All criteria/sum/average ranges cover identical source rows; text/date/number criteria match stored types; one result reconciles to a filtered/manual check.

14. Teach it back in your own words

Without the lesson open, explain sumifs, countifs and averageifs in 60-90 seconds. State the business purpose, source/result grain, one validation, one common failure and why your handling preserves meaning.

15. Protected review + changed fresh retry

Save the genuine attempt in Evidence_Log first. If repair is needed, open the protected EFA2 review, compare only the weak competency, close the review, and complete a changed retry. Append the retry; never overwrite the failed/partial attempt.

16. Evidence / mastery + exact next route

Save workbook/query output, formula/range/query evidence, independent validation or reconciliation, one failure/recovery note, confidence 1-5, and a concise own-words explanation. PASS with evidence -> continue. PARTIAL/UNCERTAIN -> targeted repair + fresh changed retry. FAIL -> repair only the named weak competency before continuing.

EFA3 - Cleaning Text and Inconsistent Values

Practice: P-EFA3_Cleaning

1. Why this lesson matters

Analyst Excel work is judged by trustworthiness and repeatability, not by how many features are used. Every transformation, formula, Pivot/query and visual must preserve meaning and survive a changed input.

2. What you are learning

Cleaning Text and Inconsistent Values at working independence: perform the analyst task, validate it, explain why the logic is safe, and recover from one realistic failure.

3. What you already need / prerequisite

M01 Excel Fundamentals plus the previous M02 lesson(s).

4. Picture it in your head

Cleaning is a controlled transformation, not permission to overwrite unusual values. Preserve raw columns, create clean helpers, and distinguish repeated values from duplicate records using the declared grain.

5. New words before we use them

TRIM = remove extra spaces. CLEAN = remove many non-printing characters. PROPER/UPPER/LOWER = case normalization. Flash Fill = inferred pattern, not a stored rule. Duplicate record = repeated entity at intended grain.

6. Watch / visual source

RECOMMENDED. Data Cleaning in Excel - 10 Tricks (Beginner to PRO) | 1:17-8:49. Why watch: See Flash Fill/Text to Columns/duplicate and cleaning patterns, then return to the safer raw-preserving helper-column workflow. Internal fallback: This lesson plus P-EFA3_Cleaning teach the controlled cleaning route.

[Open mapped source](#)

EFA3 - Follow with me once

7. Follow with me once - fully worked teacher demonstration

For raw name "riya sharma ", use =PROPER(TRIM(B2)) in a clean helper. For Customer ID duplicates, use COUNTIF against the ID column before deciding whether deletion is justified.

8. What you should see / be able to say

Raw columns remain recoverable; clean outputs are explainable; duplicate logic uses the intended key/grain; source row count changes only when a justified action is documented.

9. Why it worked

The action matches the written business meaning and keeps a validation path back to source values. The learner is not asked to trust a formula, Pivot, query, relationship or visual merely because Excel produced an output.

10. Common beginner mistakes

Editing the only raw copy; deleting a duplicate before proving the row grain; treating similar-looking labels as equivalent without evidence; using TRIM/CLEAN without checking the result.

11. If you get stuck - recovery ladder

If cleanup changes IDs or meaning, revert to raw and isolate the transformation. If Remove Duplicates seems tempting, first state exactly which column combination defines a duplicate. Recovery ladder: 1) stop editing; 2) restate the business rule/grain; 3) locate the exact formula/menu/query/relationship; 4) inspect the earliest wrong step; 5) reveal only a structural hint; 6) use protected review only after a saved attempt.

EFA3 - Do alone, validate and master

12. Now you do a small version alone

Clean a changed supplier table with messy spaces/case and repeated supplier cities. Keep raw fields and document one suspicious value that you did not automatically "fix".

13. Validate your result

Raw columns remain recoverable; clean outputs are explainable; duplicate logic uses the intended key/grain; source row count changes only when a justified action is documented.

14. Teach it back in your own words

Without the lesson open, explain cleaning text and inconsistent values in 60-90 seconds. State the business purpose, source/result grain, one validation, one common failure and why your handling preserves meaning.

15. Protected review + changed fresh retry

Save the genuine attempt in Evidence_Log first. If repair is needed, open the protected EFA3 review, compare only the weak competency, close the review, and complete a changed retry. Append the retry; never overwrite the failed/partial attempt.

16. Evidence / mastery + exact next route

Save workbook/query output, formula/range/query evidence, independent validation or reconciliation, one failure/recovery note, confidence 1-5, and a concise own-words explanation. PASS with evidence -> continue. PARTIAL/UNCERTAIN -> targeted repair + fresh changed retry. FAIL -> repair only the named weak competency before continuing.

EFA4 - Text Extraction and Date Analysis

Practice: P-EFA4_Text_Dates

1. Why this lesson matters

Analyst Excel work is judged by trustworthiness and repeatability, not by how many features are used. Every transformation, formula, Pivot/query and visual must preserve meaning and survive a changed input.

2. What you are learning

Text Extraction and Date Analysis at working independence: perform the analyst task, validate it, explain why the logic is safe, and recover from one realistic failure.

3. What you already need / prerequisite

M01 Excel Fundamentals plus the previous M02 lesson(s).

4. Picture it in your head

Extract only when the source pattern is stable. Excel dates are numeric stored values; formatting changes appearance, not the date itself. Text-looking dates must be diagnosed before arithmetic.

5. New words before we use them

LEFT/RIGHT/MID = positional extraction. LEN = length. FIND/SEARCH = locate text. YEAR/MONTH/DAY = date parts. EOMONTH = month end. Date serial = Excel numeric representation of a date.

6. Watch / visual source

OPTIONAL. Chandoo - 6 Essential TEXT functions + Date & Time values | Focused article sections; optional date companion in source registry. Why watch: Optional visual/reference reinforcement for text extraction and Excel date serial thinking. Internal fallback: This lesson plus P-EFA4_Text_Dates are the canonical teacher route.

[Open mapped source](#)

EFA4 - Follow with me once

7. Follow with me once - fully worked teacher demonstration

For INV-2026-00481, =LEFT(A2,3) returns INV. With a real Order Date, =YEAR(B2), =MONTH(B2) and =EOMONTH(B2,0) create stable date attributes.

8. What you should see / be able to say

Extraction matches the documented pattern; real dates sort/arithmetic correctly; date-looking text is not silently forced; the fixed practice date makes the result reproducible.

9. Why it worked

The action matches the written business meaning and keeps a validation path back to source values. The learner is not asked to trust a formula, Pivot, query, relationship or visual merely because Excel produced an output.

10. Common beginner mistakes

Extracting fixed character positions from unstable codes; treating a date-looking text value as a real Excel date; confusing month text labels with chronological date values.

11. If you get stuck - recovery ladder

If a positional formula returns nonsense, check whether source format varies. If date math fails, inspect stored type/value before applying formatting or DATEVALUE. Recovery ladder: 1) stop editing; 2) restate the business rule/grain; 3) locate the exact formula/menu/query/relationship; 4) inspect the earliest wrong step; 5) reveal only a structural hint; 6) use protected review only after a saved attempt.

EFA4 - Do alone, validate and master

12. Now you do a small version alone

On a changed service-code/date table, extract prefix and numeric suffix only where format is stable; create month attributes and identify one value requiring investigation before conversion.

13. Validate your result

Extraction matches the documented pattern; real dates sort/arithmetic correctly; date-looking text is not silently forced; the fixed practice date makes the result reproducible.

14. Teach it back in your own words

Without the lesson open, explain text extraction and date analysis in 60-90 seconds. State the business purpose, source/result grain, one validation, one common failure and why your handling preserves meaning.

15. Protected review + changed fresh retry

Save the genuine attempt in Evidence_Log first. If repair is needed, open the protected EFA4 review, compare only the weak competency, close the review, and complete a changed retry. Append the retry; never overwrite the failed/partial attempt.

16. Evidence / mastery + exact next route

Save workbook/query output, formula/range/query evidence, independent validation or reconciliation, one failure/recovery note, confidence 1-5, and a concise own-words explanation. PASS with evidence -> continue. PARTIAL/UNCERTAIN -> targeted repair + fresh changed retry. FAIL -> repair only the named weak competency before continuing.

EFA5 - Lookups: XLOOKUP First, Compatibility Second

Practice: P-EFA5_Lookups

1. Why this lesson matters

Analyst Excel work is judged by trustworthiness and repeatability, not by how many features are used. Every transformation, formula, Pivot/query and visual must preserve meaning and survive a changed input.

2. What you are learning

Lookups: XLOOKUP First, Compatibility Second at working independence: perform the analyst task, validate it, explain why the logic is safe, and recover from one realistic failure.

3. What you already need / prerequisite

M01 Excel Fundamentals plus the previous M02 lesson(s).

4. Picture it in your head

A lookup enriches one table from another through a key. A returned value does not prove the relationship is correct; check one-side uniqueness and keep not-found cases visible until investigated.

5. New words before we use them

Lookup value = key being searched. Lookup array = key column. Return array = field to bring back. Not Found = unmatched key. One-side uniqueness = expected one record per master key.

6. Watch / visual source

RECOMMENDED. 10 Advanced XLOOKUP Tips & Tricks | 1:15-4:18. Why watch: See XLOOKUP syntax/basic/multi-column behavior before practicing key uniqueness and Not Found controls. Internal fallback: This lesson plus Products + P-EFA5_Lookups fully teach the lookup contract.

[Open mapped source](#)

EFA5 - Follow with me once

7. Follow with me once - fully worked teacher demonstration

=XLOOKUP(B7,Products!\$A\$2:\$A\$9,Products!\$B\$2:\$B\$9,"Not Found") returns Product. Repeat for Category and Standard Price. Check Product ID uniqueness first.

8. What you should see / be able to say

Master key uniqueness is checked, not-found count is visible, enriched row count remains stable, and at least one returned value is manually verified against the master.

9. Why it worked

The action matches the written business meaning and keeps a validation path back to source values. The learner is not asked to trust a formula, Pivot, query, relationship or visual merely because Excel produced an output.

10. Common beginner mistakes

Looking up against a non-unique master key; hiding Not Found as blank; matching the wrong key column; assuming a returned value proves the correct record matched.

11. If you get stuck - recovery ladder

If values look plausible but wrong, inspect duplicate master keys and lookup/return ranges. Do not wrap unresolved #N/A in IFERROR("",...) just to make the sheet look clean. Recovery ladder: 1) stop editing; 2) restate the business rule/grain; 3) locate the exact formula/menu/query/relationship; 4) inspect the earliest wrong step; 5) reveal only a structural hint; 6) use protected review only after a saved attempt.

EFA5 - Do alone, validate and master

12. Now you do a small version alone

On a changed employee/department pair, enrich employees with department name and cost center. Deliberately include one unknown key and explain the correct data-quality response.

13. Validate your result

Master key uniqueness is checked, not-found count is visible, enriched row count remains stable, and at least one returned value is manually verified against the master.

14. Teach it back in your own words

Without the lesson open, explain lookups: xlookup first, compatibility second in 60-90 seconds. State the business purpose, source/result grain, one validation, one common failure and why your handling preserves meaning.

15. Protected review + changed fresh retry

Save the genuine attempt in Evidence_Log first. If repair is needed, open the protected EFA5 review, compare only the weak competency, close the review, and complete a changed retry. Append the retry; never overwrite the failed/partial attempt.

16. Evidence / mastery + exact next route

Save workbook/query output, formula/range/query evidence, independent validation or reconciliation, one failure/recovery note, confidence 1-5, and a concise own-words explanation. PASS with evidence -> continue. PARTIAL/UNCERTAIN -> targeted repair + fresh changed retry. FAIL -> repair only the named weak competency before continuing.

EFA6 - Error Handling without Hiding Problems

Practice: P-EFA6_Errors

1. Why this lesson matters

Analyst Excel work is judged by trustworthiness and repeatability, not by how many features are used. Every transformation, formula, Pivot/query and visual must preserve meaning and survive a changed input.

2. What you are learning

Error Handling without Hiding Problems at working independence: perform the analyst task, validate it, explain why the logic is safe, and recover from one realistic failure.

3. What you already need / prerequisite

M01 Excel Fundamentals plus the previous M02 lesson(s).

4. Picture it in your head

Errors are signals. Diagnose the cause first; only expected edge cases should receive a presentation-friendly fallback. Unexpected #N/A/#REF/#VALUE problems should remain visible until fixed.

5. New words before we use them

#N/A = no match/NA. #DIV/0! = division by zero. #VALUE! = incompatible value/type. #REF! = invalid reference. IFERROR = catch-all fallback for any error.

6. Watch / visual source

OPTIONAL. Chandoo - Excel Formula Errors: Understand and Debug Them | Focused error-type and debugging sections. Why watch: Optional reference for recognizing error types; the lesson controls when an error may be handled versus repaired. Internal fallback: This lesson plus P-EFA6_Errors are the canonical teacher route.

[Open mapped source](#)

EFA6 - Follow with me once

7. Follow with me once - fully worked teacher demonstration

Average Order Value = Sales/Orders may use a deliberate N/A/blank rule when Orders=0. A missing Product ID from XLOOKUP should normally remain visible as Not Found rather than silently blank.

8. What you should see / be able to say

Each error's cause is identified; fallbacks preserve business meaning; unresolved data-quality/reference defects remain discoverable; no catch-all hides critical problems.

9. Why it worked

The action matches the written business meaning and keeps a validation path back to source values. The learner is not asked to trust a formula, Pivot, query, relationship or visual merely because Excel produced an output.

10. Common beginner mistakes

Wrapping every formula in IFERROR; hiding a structural #REF! instead of repairing it; converting meaningful missing/unknown states to zero; not recording the original error.

11. If you get stuck - recovery ladder

If every error becomes blank, remove blanket IFERROR and inspect each case. If #REF! appears after structural edits, repair the reference rather than masking it. Recovery ladder: 1) stop editing; 2) restate the business rule/grain; 3) locate the exact formula/menu/query/relationship; 4) inspect the earliest wrong step; 5) reveal only a structural hint; 6) use protected review only after a saved attempt.

EFA6 - Do alone, validate and master

12. Now you do a small version alone

Given a changed workbook with one zero denominator, one missing lookup and one broken reference, apply three different handling choices and justify each.

13. Validate your result

Each error's cause is identified; fallbacks preserve business meaning; unresolved data-quality/reference defects remain discoverable; no catch-all hides critical problems.

14. Teach it back in your own words

Without the lesson open, explain error handling without hiding problems in 60-90 seconds. State the business purpose, source/result grain, one validation, one common failure and why your handling preserves meaning.

15. Protected review + changed fresh retry

Save the genuine attempt in Evidence_Log first. If repair is needed, open the protected EFA6 review, compare only the weak competency, close the review, and complete a changed retry. Append the retry; never overwrite the failed/partial attempt.

16. Evidence / mastery + exact next route

Save workbook/query output, formula/range/query evidence, independent validation or reconciliation, one failure/recovery note, confidence 1-5, and a concise own-words explanation. PASS with evidence -> continue. PARTIAL/UNCERTAIN -> targeted repair + fresh changed retry. FAIL -> repair only the named weak competency before continuing.

EFA7 - Data Quality Controls in Excel

Practice: P-EFA7_Quality

1. Why this lesson matters

Analyst Excel work is judged by trustworthiness and repeatability, not by how many features are used. Every transformation, formula, Pivot/query and visual must preserve meaning and survive a changed input.

2. What you are learning

Data Quality Controls in Excel at working independence: perform the analyst task, validate it, explain why the logic is safe, and recover from one realistic failure.

3. What you already need / prerequisite

M01 Excel Fundamentals plus the previous M02 lesson(s).

4. Picture it in your head

Excel controls should detect or prevent specific failures tied to grain: completeness, uniqueness, validity, consistency and reconciliation. A red cell is a signal, not automatic permission to delete/change data.

5. New words before we use them

Completeness = required value present. Uniqueness = key not repeated when it should be one-per-record. Validity = allowed domain/range. Data Validation = input-control feature. Reconciliation = compare trusted counts/totals.

6. Watch / visual source

RECOMMENDED. How to create & use Drop Down lists like a PRO - Advanced Excel Tricks | 0:00-3:58. Why watch: See dropdown creation and user guidance before building quality controls that separate prevention from detection. Internal fallback: This lesson plus P-EFA7_Quality fully teach the controls.

[Open mapped source](#)

EFA7 - Follow with me once

7. Follow with me once - fully worked teacher demonstration

If Order ID must be unique: =COUNTIF(Orders[Order ID],[@[Order ID]])>1 flags repeated IDs. Data Validation can restrict Status input to Closed/Pending. Reconcile total Net Sales to a trusted control total.

8. What you should see / be able to say

Every control has a stated business rule and grain; invalid rows are counted; input validation works; a trusted count/total reconciles before and after any justified repair.

9. Why it worked

The action matches the written business meaning and keeps a validation path back to source values. The learner is not asked to trust a formula, Pivot, query, relationship or visual merely because Excel produced an output.

10. Common beginner mistakes

Using dropdowns as if they clean old data; mixing detection and correction; deleting repeated IDs before confirming whether they should be unique; no reconciliation after changes.

11. If you get stuck - recovery ladder

If a control flags valid repeated values, revisit grain/key definition. If a dropdown breaks, inspect its source range/reference before retyping values manually. Recovery ladder: 1) stop editing; 2) restate the business rule/grain; 3) locate the exact formula/menu/query/relationship; 4) inspect the earliest wrong step; 5) reveal only a structural hint; 6) use protected review only after a saved attempt.

EFA7 - Do alone, validate and master

12. Now you do a small version alone

Build quality controls on a changed customer table: missing email, duplicate Customer ID, invalid Region and negative Lifetime Value. Keep detection separate from correction.

13. Validate your result

Every control has a stated business rule and grain; invalid rows are counted; input validation works; a trusted count/total reconciles before and after any justified repair.

14. Teach it back in your own words

Without the lesson open, explain data quality controls in excel in 60-90 seconds. State the business purpose, source/result grain, one validation, one common failure and why your handling preserves meaning.

15. Protected review + changed fresh retry

Save the genuine attempt in Evidence_Log first. If repair is needed, open the protected EFA7 review, compare only the weak competency, close the review, and complete a changed retry. Append the retry; never overwrite the failed/partial attempt.

16. Evidence / mastery + exact next route

Save workbook/query output, formula/range/query evidence, independent validation or reconciliation, one failure/recovery note, confidence 1-5, and a concise own-words explanation. PASS with evidence -> continue. PARTIAL/UNCERTAIN -> targeted repair + fresh changed retry. FAIL -> repair only the named weak competency before continuing.

EFA8 - PivotTables for Business Questions

Practice: P-EFA8_Pivots

1. Why this lesson matters

Analyst Excel work is judged by trustworthiness and repeatability, not by how many features are used. Every transformation, formula, Pivot/query and visual must preserve meaning and survive a changed input.

2. What you are learning

PivotTables for Business Questions at working independence: perform the analyst task, validate it, explain why the logic is safe, and recover from one realistic failure.

3. What you already need / prerequisite

M01 Excel Fundamentals plus the previous M02 lesson(s).

4. Picture it in your head

A PivotTable changes the result grain quickly. Rows/Columns define groups; Values define aggregation; Filters restrict population. A polished Pivot can still be wrong if source grain or aggregation is wrong.

5. New words before we use them

Rows = grouping fields. Columns = cross-grouping. Values = aggregation. Filter = restricted population. Grand total = overall aggregation. Refresh = recalculate from current source.

6. Watch / visual source

RECOMMENDED. Excel Pivot Tables - Master Data Analysis in just 45 minutes | 0:51-15:01. Why watch: See PivotTable fields, aggregation and layout before building live Pivots on Pivot_Source. Internal fallback: This lesson plus P-EFA8_Pivots + My_Pivots provide the complete internal route.

[Open mapped source](#)

EFA8 - Follow with me once

7. Follow with me once - fully worked teacher demonstration

From one-call-per-row source, Customer in Rows + Call ID Count answers calls by customer. For Value, confirm Sum; for Satisfaction, Average may be meaningful. Reconcile a grand total to source.

8. What you should see / be able to say

Correct source/Table is used, aggregation is deliberately chosen, filters are visible, refresh works, and at least one Pivot total reconciles to source.

9. Why it worked

The action matches the written business meaning and keeps a validation path back to source values. The learner is not asked to trust a formula, Pivot, query, relationship or visual merely because Excel produced an output.

10. Common beginner mistakes

Using Count when Sum/Average was intended because source values are text/blanks; building a Pivot without stating result grain; trusting a grand total without source reconciliation.

11. If you get stuck - recovery ladder

If a numeric field defaults to Count, inspect blanks/type. If totals differ, clear filters, verify source range/Table expansion and compare the aggregation/grain. Recovery ladder: 1) stop editing; 2) restate the business rule/grain; 3) locate the exact formula/menu/query/relationship; 4) inspect the earliest wrong step; 5) reveal only a structural hint; 6) use protected review only after a saved attempt.

EFA8 - Do alone, validate and master

12. Now you do a small version alone

On a changed order table, answer total sales by Region, average delivery days by Carrier and count rows by Status. Explain why Count of Order ID may not equal distinct orders.

13. Validate your result

Correct source/Table is used, aggregation is deliberately chosen, filters are visible, refresh works, and at least one Pivot total reconciles to source.

14. Teach it back in your own words

Without the lesson open, explain pivottables for business questions in 60-90 seconds. State the business purpose, source/result grain, one validation, one common failure and why your handling preserves meaning.

15. Protected review + changed fresh retry

Save the genuine attempt in Evidence_Log first. If repair is needed, open the protected EFA8 review, compare only the weak competency, close the review, and complete a changed retry. Append the retry; never overwrite the failed/partial attempt.

16. Evidence / mastery + exact next route

Save workbook/query output, formula/range/query evidence, independent validation or reconciliation, one failure/recovery note, confidence 1-5, and a concise own-words explanation. PASS with evidence -> continue. PARTIAL/UNCERTAIN -> targeted repair + fresh changed retry. FAIL -> repair only the named weak competency before continuing.

EFA9 - PivotCharts, Slicers and Report Design

Practice: P-EFA9_Report

1. Why this lesson matters

Analyst Excel work is judged by trustworthiness and repeatability, not by how many features are used. Every transformation, formula, Pivot/query and visual must preserve meaning and survive a changed input.

2. What you are learning

PivotCharts, Slicers and Report Design at working independence: perform the analyst task, validate it, explain why the logic is safe, and recover from one realistic failure.

3. What you already need / prerequisite

M01 Excel Fundamentals plus the previous M02 lesson(s).

4. Picture it in your head

A report is a decision-support view, not a collection of charts. Build and validate the Pivot first; then choose a visual fitting the relationship and a slicer only when users genuinely need that interaction.

5. New words before we use them

PivotChart = chart tied to PivotTable. Slicer = visible filter. Report Connection = slicer connection to one/more Pivots. Time trend = ordered date/month. Category comparison = independent groups.

6. Watch / visual source

RECOMMENDED. Excel Pivot Tables - Master Data Analysis in just 45 minutes | 20:32-24:40. Why watch: See PivotChart/slicer reporting patterns before building My_Report and testing filter/clear states. Internal fallback: This lesson plus P-EFA9_Report + My_Report fully teach the reporting route.

[Open mapped source](#)

EFA9 - Follow with me once

7. Follow with me once - fully worked teacher demonstration

Create monthly call-count PivotTable, add a line PivotChart, add Representative slicer, test default/filter/clear states and confirm chart/Pivot totals stay consistent.

8. What you should see / be able to say

Chart type fits question, slicer connections are correct, KPI/Pivot/chart remain consistent through filter states, titles are meaningful and no visual exaggeration/3D is used.

9. Why it worked

The action matches the written business meaning and keeps a validation path back to source values. The learner is not asked to trust a formula, Pivot, query, relationship or visual merely because Excel produced an output.

10. Common beginner mistakes

Too many slicers/visuals; using a line chart for unordered categories; forgetting slicer Report Connections; writing causal claims from descriptive data.

11. If you get stuck - recovery ladder

If visuals disagree, validate the underlying Pivot before formatting. If slicer affects the wrong view, inspect Report Connections. If a sentence claims cause, rewrite as observed pattern. Recovery ladder: 1) stop editing; 2) restate the business rule/grain; 3) locate the exact formula/menu/query/relationship; 4) inspect the earliest wrong step; 5) reveal only a structural hint; 6) use protected review only after a saved attempt.

EFA9 - Do alone, validate and master

12. Now you do a small version alone

Build a changed report with one category comparison and one ordered time trend. Use only one useful slicer and write two evidence-based observations without claiming causation.

13. Validate your result

Chart type fits question, slicer connections are correct, KPI/Pivot/chart remain consistent through filter states, titles are meaningful and no visual exaggeration/3D is used.

14. Teach it back in your own words

Without the lesson open, explain pivotcharts, slicers and report design in 60-90 seconds. State the business purpose, source/result grain, one validation, one common failure and why your handling preserves meaning.

15. Protected review + changed fresh retry

Save the genuine attempt in Evidence_Log first. If repair is needed, open the protected EFA9 review, compare only the weak competency, close the review, and complete a changed retry. Append the retry; never overwrite the failed/partial attempt.

16. Evidence / mastery + exact next route

Save workbook/query output, formula/range/query evidence, independent validation or reconciliation, one failure/recovery note, confidence 1-5, and a concise own-words explanation. PASS with evidence -> continue. PARTIAL/UNCERTAIN -> targeted repair + fresh changed retry. FAIL -> repair only the named weak competency before continuing.

EFA10 - Power Query: Import, Transform, Load and Refresh

Practice: P-EFA10_PQ

1. Why this lesson matters

Analyst Excel work is judged by trustworthiness and repeatability, not by how many features are used. Every transformation, formula, Pivot/query and visual must preserve meaning and survive a changed input.

2. What you are learning

Power Query: Import, Transform, Load and Refresh at working independence: perform the analyst task, validate it, explain why the logic is safe, and recover from one realistic failure.

3. What you already need / prerequisite

M01 Excel Fundamentals plus the previous M02 lesson(s).

4. Picture it in your head

Power Query stores repeatable transformation steps: Connect -> Inspect -> Transform -> Validate -> Load -> Refresh. It should replace repeated manual cleanup, not hide unexplained row/total changes.

5. New words before we use them

Query = stored transformation. Applied Steps = ordered transformations. Data type = Text/Date/Number/etc. Load = write query output. Refresh = rerun steps on current source.

6. Watch / visual source

RECOMMENDED. Power Query - Beginner to PRO Masterclass in 30 minutes | 4:54-13:36; 17:50-28:25; 33:35-35:17. Why watch: See Editor, cleaning/loading, staff/date transformations and refresh before reproducing the controlled PQ_Source workflow. Internal fallback: This lesson plus P-EFA10_PQ give numbered click-by-click instructions and a refresh test.

[Open mapped source](#)

EFA10 - Follow with me once

7. Follow with me once - fully worked teacher demonstration

Load a source range, promote headers if needed, set types deliberately, Trim/Clean text, replace a justified known value, split/filter/add conditional column, then Close & Load and Refresh after a source change.

8. What you should see / be able to say

Applied Steps are readable, types are deliberate, row-count/total changes are explained, raw source remains untouched, refresh updates automatically and output reconciles where definitions match.

9. Why it worked

The action matches the written business meaning and keeps a validation path back to source values. The learner is not asked to trust a formula, Pivot, query, relationship or visual merely because Excel produced an output.

10. Common beginner mistakes

Manually editing the loaded output instead of fixing the query; deleting errors without diagnosing them; changing type before inspecting bad values; refresh depends on fragile source assumptions.

11. If you get stuck - recovery ladder

If a later step breaks, locate the first failing Applied Step and inspect renamed/removed columns or type assumptions. If row count changes unexpectedly, stop and explain before continuing. Recovery ladder: 1) stop editing; 2) restate the business rule/grain; 3) locate the exact formula/menu/query/relationship; 4) inspect the earliest wrong step; 5) reveal only a structural hint; 6) use protected review only after a saved attempt.

EFA10 - Do alone, validate and master

12. Now you do a small version alone

On a changed file/range, build a query that cleans text, sets types, flags or filters one documented issue, loads a curated table and passes a refresh-change test.

13. Validate your result

Applied Steps are readable, types are deliberate, row-count/total changes are explained, raw source remains untouched, refresh updates automatically and output reconciles where definitions match.

14. Teach it back in your own words

Without the lesson open, explain power query: import, transform, load and refresh in 60-90 seconds. State the business purpose, source/result grain, one validation, one common failure and why your handling preserves meaning.

15. Protected review + changed fresh retry

Save the genuine attempt in Evidence_Log first. If repair is needed, open the protected EFA10 review, compare only the weak competency, close the review, and complete a changed retry. Append the retry; never overwrite the failed/partial attempt.

16. Evidence / mastery + exact next route

Save workbook/query output, formula/range/query evidence, independent validation or reconciliation, one failure/recovery note, confidence 1-5, and a concise own-words explanation. PASS with evidence -> continue. PARTIAL/UNCERTAIN -> targeted repair + fresh changed retry. FAIL -> repair only the named weak competency before continuing.

EFA11 - Relationships and the Excel Data Model (Strong Fresher)

Practice: P-EFA11_Model

1. Why this lesson matters

Analyst Excel work is judged by trustworthiness and repeatability, not by how many features are used. Every transformation, formula, Pivot/query and visual must preserve meaning and survive a changed input.

2. What you are learning

Relationships and the Excel Data Model (Strong Fresher) at working independence: perform the analyst task, validate it, explain why the logic is safe, and recover from one realistic failure.

3. What you already need / prerequisite

M01 Excel Fundamentals plus the previous M02 lesson(s).

4. Picture it in your head

Fact-like event tables often contain repeated foreign keys; dimension-like tables should usually contain one row per key. A one-to-many relationship depends on uniqueness on the one side.

5. New words before we use them

Fact-like table = repeated transactions/events. Dimension-like table = descriptive entities. Relationship = key-based connection. Cardinality = matching multiplicity. One side = unique key side. Many side = repeating foreign key side.

6. Watch / visual source

RECOMMENDED_REFERENCE. Microsoft Support - Create a relationship between tables in Excel | Matching keys, uniqueness, one-to-many, create/manage relationships. Why watch: Use the official Excel relationship explanation after grain/key uniqueness is understood; feature availability can vary by Excel edition. Internal fallback: This lesson plus Model_Products/Model_Sales + P-EFA11_Model teach the grain/cardinality reasoning even if the relationship UI is unavailable.

[Open mapped source](#)

EFA11 - Follow with me once

7. Follow with me once - fully worked teacher demonstration

Sales contains many P104 rows; Products should contain one P104. Product ID connects many Sales rows to one Products row. Two P104 rows in Products break the intended one-side uniqueness.

8. What you should see / be able to say

Grain is stated for both tables, intended one-side key is unique, relationship direction/cardinality is correctly explained, and no DAX mastery is assumed.

9. Why it worked

The action matches the written business meaning and keeps a validation path back to source values. The learner is not asked to trust a formula, Pivot, query, relationship or visual merely because Excel produced an output.

10. Common beginner mistakes

Creating a relationship before proving the one-side key is unique; confusing many-to-one direction; solving duplicate dimension keys with arbitrary deletion; assuming UI availability equals conceptual mastery.

11. If you get stuck - recovery ladder

If relationship creation fails, inspect data types, blanks and duplicate keys on the one side. Do not “fix” duplicate dimension keys by arbitrary deletion without investigating entity meaning. Recovery ladder: 1) stop editing; 2) restate the business rule/grain; 3) locate the exact formula/menu/query/relationship; 4) inspect the earliest wrong step; 5) reveal only a structural hint; 6) use protected review only after a saved attempt.

EFA11 - Do alone, validate and master

12. Now you do a small version alone

Given a changed Orders/Customers pair, identify many/one side, validate Customer ID uniqueness on Customers and predict the defect caused by duplicate customer keys.

13. Validate your result

Grain is stated for both tables, intended one-side key is unique, relationship direction/cardinality is correctly explained, and no DAX mastery is assumed.

14. Teach it back in your own words

Without the lesson open, explain relationships and the excel data model (strong fresher) in 60-90 seconds. State the business purpose, source/result grain, one validation, one common failure and why your handling preserves meaning.

15. Protected review + changed fresh retry

Save the genuine attempt in Evidence_Log first. If repair is needed, open the protected EFA11 review, compare only the weak competency, close the review, and complete a changed retry. Append the retry; never overwrite the failed/partial attempt.

16. Evidence / mastery + exact next route

Save workbook/query output, formula/range/query evidence, independent validation or reconciliation, one failure/recovery note, confidence 1-5, and a concise own-words explanation. PASS with evidence -> continue. PARTIAL/UNCERTAIN -> targeted repair + fresh changed retry. FAIL -> repair only the named weak competency before continuing.

M-XLA - Build an Excel Analysis That Can Be Defended

1. Why this project matters

A junior analyst is hired to turn messy business tables into a decision-ready answer, not to demonstrate isolated formulas. This project joins the skills you already learned into one auditable workflow.

2. What you are building

Take unfamiliar supplied tables from raw inspection to a validated management summary and defend formulas, transformations, Pivots, visuals, findings and limitations.

3. Prerequisite

EFA1-EFA11 practiced with fresh evidence. Power Query and relationship/cardinality concepts must already have been taught before this project begins.

4. Picture it in your head

Raw source -> quality log -> prepared/enriched data -> calculations -> Pivots -> management report -> independent reconciliations -> findings/recommendation/limitations. If the report is wrong, trace backward to the earliest wrong layer.

5. New words before we use them

Prepared data = working copy after documented transformations. Reconciliation = independent check that two ways of computing an important control agree. Management report = concise decision-facing output. Limitation = what the current evidence cannot safely prove.

6. Watch / visual source

INTERNAL PROJECT TEACHER. No new external video is required; every project skill has already been taught in EFA1-EFA11. Rewatch a lesson-specific visual only for a named weak skill, not as project binge-watching.

M-XLA - Beginner setup and worked micro-example

7. Follow with me once - worked micro-example

Open Project_Sales. Do not edit it. In Validation_Log write the raw grain: one row per order line. Copy/work into Prepared_Data. For one valid row calculate Net Sales = Quantity * Unit Price. Then classify delivery: cancelled -> Excluded; otherwise Delivered Date <= Promised Date -> On Time; else Late. Manually recompute that same row and verify Product ID against Project_Products.

8. What you should see / be able to say

You can point to the untouched raw row, the prepared row, the calculation, the lookup result and the manual check. Unknown product/target keys remain visible rather than silently disappearing.

9. Why it worked

One tiny row proves the full reasoning chain before you scale it to all rows. It separates source evidence, transformation, enrichment, calculation and validation.

10. Common beginner mistakes

Building the dashboard before defining metrics; editing Project_Sales; hiding Not Found values; using a lookup against a non-unique master key; deleting rows without a rejection rule; trusting a Pivot total without reconciliation.

11. If you get stuck - recovery ladder

If the report looks finished but cannot be defended, trace backward: visual -> Pivot -> prepared metric -> transformation -> raw grain. Repair the earliest incorrect step and rerun validations. Stop at the earliest wrong layer and repair only that layer before rebuilding downstream outputs.

M-XLA - Full independent build

12. Now you do the project

- Preserve Project_Sales / Project_Products / Project_Targets untouched.
- In Validation_Log write source grain, business questions, metric definitions and quality/rejection rules.
- Build Prepared_Data and document every cleaning/standardization decision.
- Create Net Sales and Delivery Result; use IF/IFS/logic deliberately and test boundaries/missing cases.
- Enrich Product/Category/target information only after proving the master key is suitable; keep unknown keys visible.
- Use conditional aggregation where it gives a reusable control/report answer.
- Build monthly, region, category and delivery-performance Pivot analyses in Analysis_Pivots.
- Create concise KPIs and only 1-2 useful visuals in Management_Report; add a slicer only when it helps a decision.
- Use Power Query for at least one refreshable preparation route where assigned; document source/output row counts.
- Reconcile at least two important totals/counts independently. Write three supported findings, one recommendation and explicit limitations.

13. Validate

Required: one manual row calculation + one row/key/not-found control + at least two reconciliations. Report metrics must use the same definitions as the prepared data/Pivots.

If any control fails, do not explain around it; repair the earliest wrong step.

M-XLA - Explain, evidence and project rubric

14. Teach it back in your own words

Give a 2-minute defense: business question; raw and result grain; one important transformation; lookup/master-key reasoning; metric definitions; two validation controls; one failure/recovery; one limitation; and the decision your strongest finding supports.

15. Protected review + changed retry

Save the complete project attempt first. If a rubric competency is weak, use only the relevant lesson/project review to repair it. Close review material and rebuild the affected component on changed data before marking the project mastered.

16. Evidence / mastery + exact next route

Critical pass requirements: raw preserved; grain/definitions explicit; transformations justified; lookup/not-found controls visible; formulas/Pivots/report agree; refresh/repeatability evidence where assigned; at least two independent reconciliations; findings supported; limitation stated; learner can rebuild and defend without answer material. A polished dashboard cannot compensate for a critical data/logic failure. PASS -> Gate A. PARTIAL/FAIL -> targeted repair only, then changed retry before Gate A.

End-of-module mastery checklist

- EFA1: Translate business rules into IF/IFS/AND/OR logic and test boundaries/blanks.
- EFA2: Build cell-driven SUMIFS/COUNTIFS/AVERAGEIFS reports and reconcile one result.
- EFA3: Clean text/inconsistency while preserving raw and documenting assumptions.
- EFA4: Extract stable text components and create reliable date analysis fields.
- EFA5: Use XLOOKUP only after checking master-key suitability; keep Not Found visible.
- EFA6: Handle expected errors without hiding structural/data-quality problems.
- EFA7: Build prevention/detection/reconciliation controls and keep correction separate.
- EFA8: Build/reconcile PivotTables and explain the summary grain/aggregation.
- EFA9: Build clear PivotCharts/slicer report and test default/filter/clear states.
- EFA10: Build a repeatable Power Query preparation and prove refresh on a controlled source change.
- EFA11: State table grain, prove one-side key uniqueness and explain one-to-many relationship/cardinality.
- M-XLA: Defend a full Excel analyst workflow with evidence, validations, findings and limitations.

Gate readiness

Attempt Gate A only when every critical skill has fresh evidence and M-XLA is defensible.

A clean Gate A pass ends the Gate. B/C are remediation retests, never normal progression.

Revision / quick reference

IF / IFS / AND / OR	Write the business rule first; test boundaries and blanks.
SUMIFS / COUNTIFS / AVERAGEIFS	Correct ranges + correct stored types + cell-driven criteria + reconciliation.
Cleaning	Preserve raw; helper/prepared fields; diagnose before changing/deleting.
Text / dates	Extract only stable patterns; a date-looking text value is not automatically a true date.
XLOOKUP	Check lookup-key uniqueness; keep Not Found visible; manually verify one match.
Errors	Expected error may be handled; structural/data-quality error must be repaired or remain visible.
Quality controls	Prevention != detection != correction; reconcile after changes.
PivotTables	State summary grain and aggregation; reconcile one important total.
Pivot report	Choose visuals by question; test slicer states; do not invent causes.
Power Query	Inspect -> transform -> load -> refresh -> validate; never hand-fix the loaded output.
Relationships	One side must be appropriate/unique; define grain/cardinality before clicking.

Next route: M-XLA -> Gate A -> M03 SQL Foundations only after M02 competency pass.